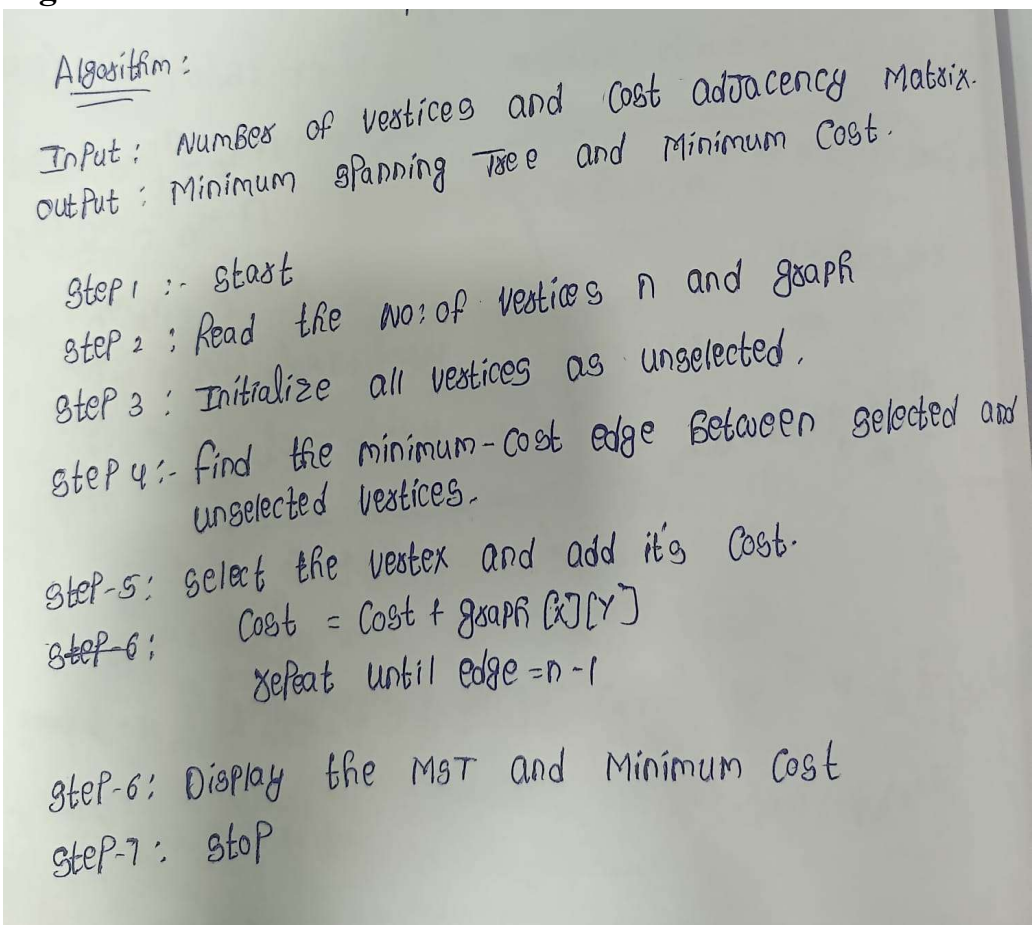
practical-9

Aim: To Implement prim's algorithm

prim's algorithm

Algorithm:



Program:

```
#include <iostream> using
namespace std; int main()
{    int
graph[MAX][MAX];    int
selected[MAX];
```

```
    int n, i, j;     int
edge = 0;     int x,
y;     int min, cost =
0;

    cout << "Enter number of vertices: ";
cin >> n;

    cout << "Enter Cost Adjacency Matrix:\n";

    for(i = 0; i < n; i++)
    {         for(j = 0; j < n;
j++)
        {             cin >>
graph[i][j];

            if(graph[i][j] == 0)
graph[i][j] = INF;
        }
    }

    // Initially select first vertex
for(i = 0; i < n; i++)
selected[i] = 0;

    selected[0] = 1;

    cout << "\nEdges in Minimum Spanning Tree:\n";

    while(edge < n - 1)
    {         min =
INF;         for(i
= 0; i < n; i++)
        {
if(selected[i])
            {                 for(j = 0; j
< n; j++)
                {
                    if(!selected[j] && graph[i][j] < min)
```

```
                {
min = graph[i][j];
x = i;                  y = j;
                }
            }
        }
    }

    cout << x << " --> " << y << "  Cost = " << graph[x][y] << endl;

    cost += graph[x][y];
selected[y] = 1;        edge++;
  }

  cout << "\nMinimum Cost = " << cost << endl;

  return 0;
}
```

Output:

```
Output

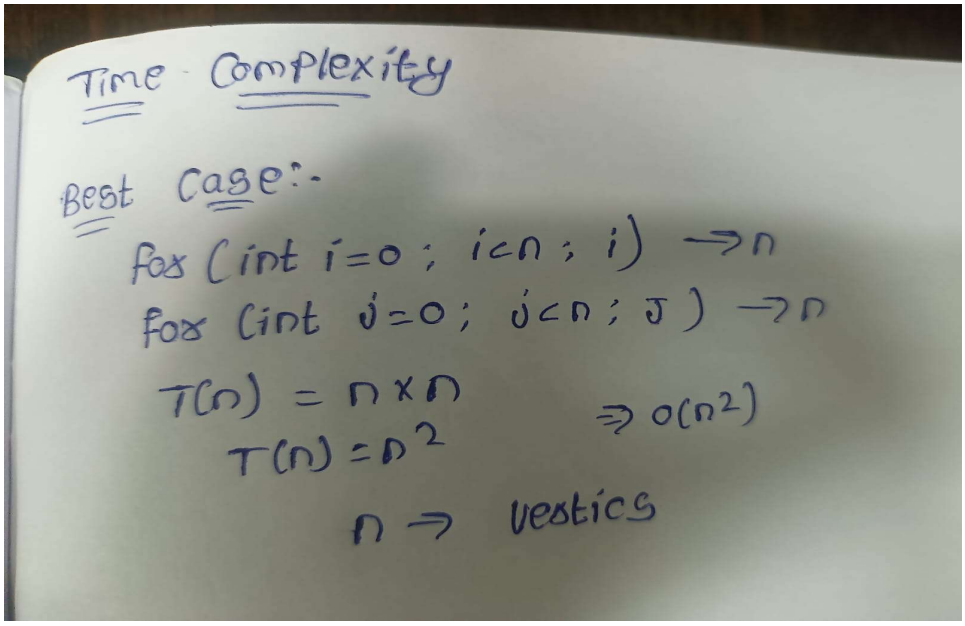
Enter number of vertices: 6
Enter Cost Adjacency Matrix:
0 2 3 0 0 0
2 0 1 4 0 0
3 1 0 5 6 0
0 4 5 0 2 3
0 0 6 2 0 1
0 0 0 3 1 0

Edges in Minimum Spanning Tree:
0 --> 1   Cost = 2
1 --> 2   Cost = 1
1 --> 3   Cost = 4
3 --> 4   Cost = 2
4 --> 5   Cost = 1

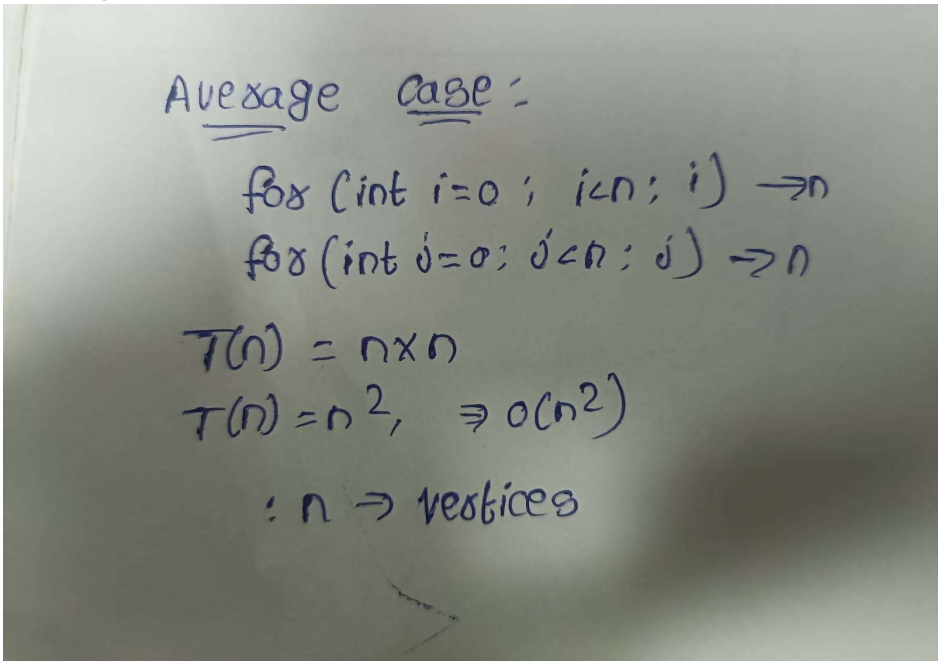
Minimum Cost = 10
```

Time Complexity:

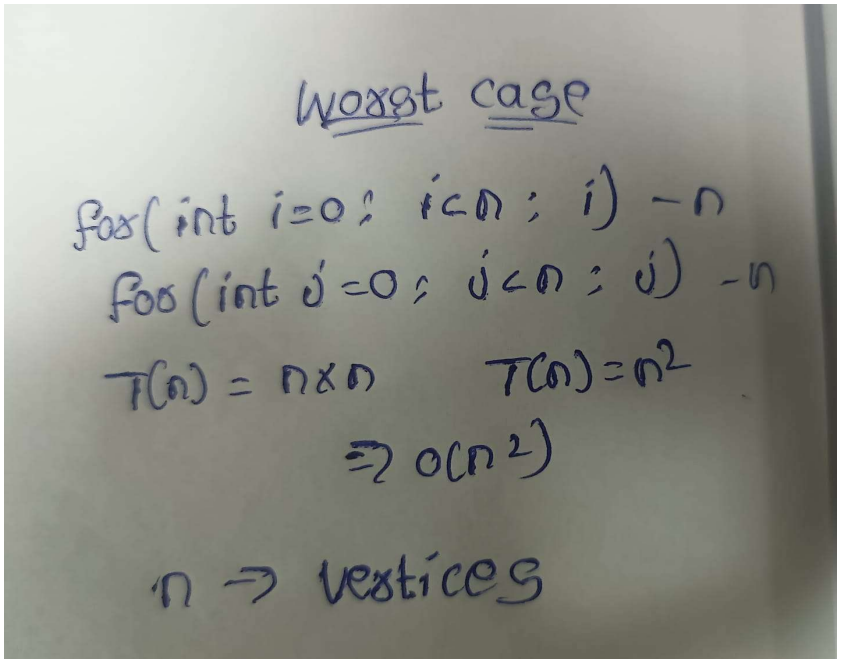
Best Case:



Average Case:



Worst Case:



Conclusion:

The implementation of **Prim's Algorithm** helps us understand how a **Minimum Spanning Tree (MST)** is constructed for a weighted graph. It repeatedly selects the minimum-cost edge connecting a selected vertex to an unselected vertex. For the given adjacency matrix implementation, Prim's Algorithm has a time complexity of $\mathbf{O(n^2)}$ in the best, average, and worst cases. Therefore, Prim's Algorithm is a useful technique for finding the minimum-cost spanning tree of a connected graph.